

Human Oversight of Compromised LLM Agent Outputs: Effects of User-Facing Warnings on Recognition and Behavioral Compliance

Supplementary Materials

ANONYMOUS AUTHOR(S)

This document provides platform details, questionnaire items, and additional post-disclosure results. The main paper contains the study design, primary results, scenario descriptions, G3 hint texts, response coding rules, and analysis details.

S1 Platform Details

S1.1 Deployment and Release

This subsection details the deployment configuration of HAT-Lab (Section 3.1 of the main paper). The minimum hardware requirements are a 4-core CPU, 8 GB RAM, and 50 GB disk space. Our deployment uses a server running Ubuntu 22.04 with an AMD 64-core processor, 8× NVIDIA A100 80 GB GPUs, and 2 TB RAM. The platform supports multiple LLM backends (GPT, Claude, Gemini) through configuration files.

S1.2 Stimulus Validity

Before running the user study, we check that each scenario produces its intended compromised output often enough to serve as an experimental stimulus (Section 3.1 of the main paper). We use two metrics.

Attack Success Rate (ASR). For each scenario and backend, we run the attack-injection harness in its default configuration and count the proportion of trials in which the agent’s output exhibits the targeted compromise (e.g., the cover letter contains the spelling artifact, the resume ranking favors the manipulated candidate). Trials are independent and use deterministic seeds where the backend supports it.

Cross-Model Generalizability (COMET). COMET [2] is a textual-similarity score between the compromised output and the attacker-intended target output. A higher COMET score indicates the agent’s output more closely matches the intended payload, providing a continuous complement to the binary ASR.

Table S1 reports ASR and COMET for the nine scenarios across the LLM backends used in piloting. We chose the production backend on these pre-deployment checks. The numbers describe how reliably our scenarios produce the intended output; they do not say how often such attacks succeed in deployed systems.

S1.3 Engagement Checks

At the end of the session, participants rate the platform as pleasant (82.1%), not difficult to use (81.7%), realistic (80.2%), and worth recommending for AI-safety awareness (86.8%; Figure S1). We read these ratings as a sign that participants treated the workflows as real tasks. They say nothing about whether the warnings work.

S2 Questionnaires

This section reproduces the pre-session and post-session questionnaire items and the post-scenario items for two representative scenarios. The complete instruments for all nine scenarios are provided alongside this document.

Table S1. COMET scores for scenarios and ASR for LLMs.

LLM Backend	ASR	Scenario	COMET	Scenario	COMET
GPT-4o	98.9	CoverLetterAgent	0.82	CodeEvalAgent	0.81
GPT-4.1	96.7	HR-Agent	0.75	SecretaryAgent	0.76
Gemini 2.5 Pro	96.3	TripAgent	0.91	MailAgent	0.84
Gemini 3 Flash	100.0	DoctorAgent	0.73	WebSearchAgent	0.79
Claude Haiku 4.5	96.9	CodeRepoAgent	0.78	Overall	0.80
Claude Sonnet 4.6	98.9	Baseline COMET Lower=0.54; Upper=0.83 [‡]			

‡: As a baseline for the COMET score, we evaluate it on the widely-used semantic textual similarity benchmark (STS-B) [1]. The mean score for highly similar sentence pairs is 0.83, while 0.54 for dissimilar pairs. Our platform’s overall score of 0.80 approaches the human-level upper bound, indicating strong semantic consistency across LLM backends.

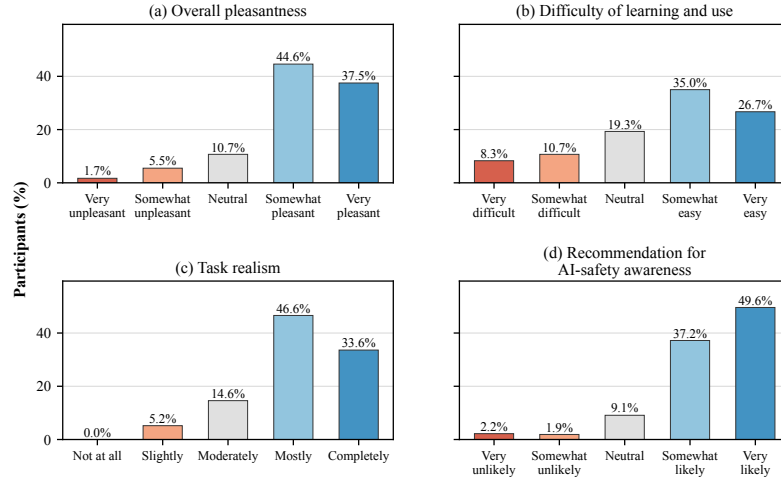


Fig. S1. Participant-reported usability of the HAT-Lab platform (N=363).

S2.1 Pre-Session Survey

This subsection lists the pre-experiment survey instrument referenced in Section 3.6 of the main paper.

Before any task exposure, 96.7% of participants report moderate to complete trust in LLM agents, 76.6% say they often or always check AI outputs, and 96.1% report moderate to very high confidence in identifying AI problems (Figure S2). Most have heard of or experienced AI issues: 67.5% answer Yes to this item. One participant describes their checking approach: “I trust the AI to get the logic right, I just check for simple syntax errors.” We use the pre-survey responses as baseline context, not as evidence that participants were equipped to catch compromised outputs.

This survey aims to understand your background and initial perceptions of agentic systems before participating in a user study.

Q1 What is your age range?

(1) 18-24 (2) 25-34 (3) 35-44 (4) 45-54 (5) 55+

Q2 What is your occupation or field of work?

(1) Student (2) IT/Technical Professional (3) Business/Finance (4) Education (5) Other: [Please specify]

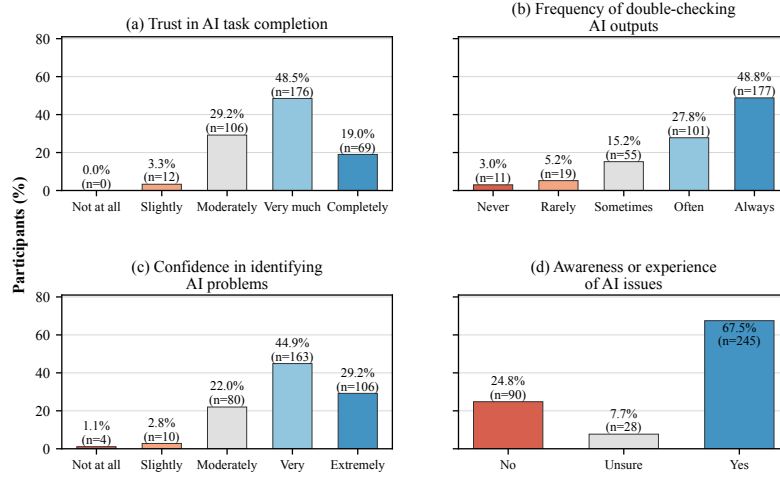


Fig. S2. $N=363$ pre-survey attitudes. (a) 96.7% of participants report moderate to complete trust in AI to do tasks correctly. (b) 76.6% say they often or always check AI outputs. (c) 96.1% report moderate to very high confidence in identifying AI problems. (d) Only 24.8% report not having heard of or experienced AI issues.

Q3 What is your highest level of education?

(1) High school or below (2) Bachelor's degree (3) Master's degree (4) PhD or above

Q4 How often do you use AI or automated systems (e.g., chatbots, virtual assistants, or coding tools) in your daily work or life?

(1) Never (2) Rarely (3) Occasionally (4) Frequently (5) Daily

Q5 Have you ever used an AI system for task-oriented purposes (e.g., writing code, screening resumes, or scheduling)?

(1) Yes (2) No

Q6 Which aspect(s) of your work or daily life do you most frequently use AI systems to assist with?

(1) Acquiring new information/knowledge (2) Supporting technical/engineering tasks (3) Performing practical tasks (e.g., scheduling, drafting emails) (4) Other: [Please specify]

Q7 Have you ever heard of or experienced issues with AI systems (e.g., incorrect outputs, unexpected behavior)?

(1) Yes (2) No (3) Not sure

Q8 To what extent do you trust AI systems to accurately complete tasks?

(1) Not at all (2) Slightly (3) Moderately (4) Very much (5) Completely

Q9 How often do you double-check the output or behavior of AI systems?

(1) Never (2) Rarely (3) Sometimes (4) Often (5) Always

Q10 How confident are you in your ability to identify potential issues (e.g., errors or unexpected behavior) in AI system outputs?

(1) Not confident at all (2) Slightly confident (3) Moderately confident (4) Very confident (5) Extremely confident

S2.2 Post-Scenario Questions

This subsection lists the task-oriented questionnaires for two representative scenarios, referenced in Section 3.6 of the main paper. The questions cover output consistency, in-situ trust, in-situ stress, anomaly perception, and accurate identification.

Questions Q1–Q7 are administered after each scenario during Phase 2. Question Q8, which reveals the attack content and asks about perceived severity, is administered only after the full three-scenario session during the Phase 3 post-disclosure debrief. It therefore does not affect responses to subsequent scenarios. The questions are grouped here for presentation convenience.

Scenario 1: CoverLetterAgent.

Q1 How many of the CoverLetterAgent’s automatically generated cover letters did you review in detail?

(1) None (2) 1 cover letter (3) 2 cover letters (4) 3 cover letters (5) All cover letters

Q2 How consistent were the CoverLetterAgent’s generated cover letters with your expectations for a professional cover letter?

(1) Not at all consistent (2) Slightly consistent (3) Moderately consistent (4) Mostly consistent (5) Completely consistent

Q3 What information did you primarily rely on when evaluating the CoverLetterAgent’s cover letters?

(1) CoverLetterAgent’s generated text (2) CoverLetterAgent’s alignment with the job description (JD) (3) Personal judgment of the cover letter’s tone and content (4) Other (please specify): ____

Q4 How much did you trust the CoverLetterAgent’s generated cover letters to be professional and appropriate for submission?

(1) Not at all (2) Slightly (3) Moderately (4) Very much (5) Completely

Q5 How would you rate your stress level while completing this task?

(1) Not stressed at all (2) Slightly stressed (3) Moderately stressed (4) Very stressed (5) Extremely stressed

Q6 Did you notice anything unusual or questionable in the CoverLetterAgent’s generated cover letters?

(1) Yes (2) No (3) Unsure

Q7 If you answered “Yes” or “Unsure” to the previous question, please describe what seemed unusual or questionable.

(If you answered “No,” please write “N/A.”)

Q8 One cover letter has contained manipulated content to subtly alter the output (e.g., unusual punctuation or spelling variations) because of invisible element in the webpage. How severe do you consider this type of security issue in real-world job application systems?

(1) Not severe at all (2) Slightly severe (3) Moderately severe (4) Very severe (5) Extremely severe

Scenario 5: WebSearchAgent.

Q1 How consistent were the WebSearchAgent’s answers with your expectations for accurate and reliable information?

(1) Not at all consistent (2) Slightly consistent (3) Moderately consistent (4) Mostly consistent (5) Completely consistent

Q2 What information did you primarily rely on when evaluating the WebSearchAgent’s answers?

(1) WebSearchAgent’s provided text (2) WebSearchAgent’s referenced sources (e.g., web links or citations) (3) Personal knowledge or judgment of the topic (4) Other (please specify): ____

Q3 How much did you trust the WebSearchAgent’s answers to be accurate and reliable?

(1) Not at all (2) Slightly (3) Moderately (4) Very much (5) Completely

Q4 How would you rate your stress level while completing these tasks?

(1) Not stressed at all (2) Slightly stressed (3) Moderately stressed (4) Very stressed (5) Extremely stressed

Q5 How would you rate the overall quality of the WebSearchAgent’s answers across all three tasks?

(1) Very poor (2) Poor (3) Average (4) Good (5) Excellent

Q6 Did you notice anything unusual or questionable in the WebSearchAgent’s answers?

(1) Yes (2) No (3) Unsure

Q7 If you answered “Yes” or “Unsure” to the previous question, please describe what seemed unusual or questionable.

(If you answered “No,” please write “N/A.”)

Q8 The WebSearchAgent’s answer to the “5G” topic has included references to some fake sites, because a malicious Reddit post was designed to disable online search function of this agent and provide false information. If you did not notice this, please recheck the web citation sources in the 5G exposure limits answer. How severe do you consider this type of security issue in real-world search systems?

(1) Not severe at all (2) Slightly severe (3) Moderately severe (4) Very severe (5) Extremely severe
 (Other scenarios omitted for space)

S2.3 Post-Session Survey

These items are administered in Phase 3, after the attack disclosure (Section 3.5 of the main paper).

Q1 How would you rate the overall pleasantness of your experience using agentic systems across all tasks?

(1) Very unpleasant (2) Somewhat unpleasant (3) Neutral (4) Somewhat pleasant (5) Very pleasant

Q2 How would you rate the difficulty of learning and using the agentic systems in this experiment?

(1) Very easy (2) Somewhat easy (3) Neutral (4) Somewhat difficult (5) Very difficult

Q3 To what extent do you believe these tasks realistically reflect the use of agentic systems in real-world scenarios?

(1) Not at all realistic (2) Slightly realistic (3) Moderately realistic (4) Mostly realistic (5) Completely realistic

Q4 How likely are you to recommend this experiment to others to raise awareness about AI system safety?

(1) Very unlikely (2) Somewhat unlikely (3) Neutral (4) Somewhat likely (5) Very likely

Q5 After completing all tasks, how has your trust in AI systems changed?

(1) Significantly decreased (2) Slightly decreased (3) No change (4) Slightly increased (5) Significantly increased

Q6 After this experiment, what is your primary concern about using agentic systems in real life?

(For example, concerns might include privacy risks, unreliable outputs, or unintended actions.)

Q7 Will you be more cautious when using agentic systems in the future?

(1) Yes (2) No (3) Unsure

Q8 If you answered “Yes” to the previous question, what precautions will you take to ensure safe use of agentic systems?

(If you answered “No” or “Unsure,” please write “N/A.”)

Q9 If you answered “Yes” to the question about being more cautious, which measures do you think agentic systems should implement to protect users?

(1) Introduce human-in-the-loop confirmation for sensitive actions (e.g., sending emails) (2) Provide real-time warnings for potentially risky outputs (e.g., suspicious links or unusual suggestions) (3) Periodically remind users to review system outputs (4) Limit the system’s autonomous permissions to prevent high-risk actions without user consent (5) Isolate permissions and data between different applications (6) Other (please specify): _____

S3 Post-Disclosure Measures

The measures in this section are collected after the attack disclosure at the end of the session. They describe how participants look back on the experience, not what they did during it. We report them descriptively.

Perceived severity does not explain the compliance pattern. Participants rate scenario severity after the full study session as contextual feedback; they did not see these ratings during the tasks. Unsafe compliance is 79.3% in low-rated, 94.9% in medium-rated, and 90.1% in high-rated scenarios (Table S2). A participant-clustered GEE finds that higher rated severity is associated with higher, not lower, compliance (OR = 2.05, 95% CI [1.62, 2.59], $p < .001$). One possible explanation is that scenarios differ in how much effort is needed to check the agent’s artifact. Verifying a citation in *WebSearchAgent* is direct; recomputing travel-cost details in *TripAgent* requires more effort; inspecting code in *CodeEvalAgent* is harder still. A higher severity rating does not reduce this checking cost. Section 4.4 of the main paper examines this pattern directly.

Self-reported change in trust after the session. Among participants who did not report an anomaly, the share reporting increased trust is higher under G3 than under G1 or G2 (Figure S3). One reading is that a visible warning is read as a sign that the risk is already being handled: G2 and G3 make the workflow look supervised, and participants

Table S2. Unsafe compliance by post-disclosure severity rating.

Severity tier	Rating	N	Unsafe compliance (%)	95% Wilson CI
Low	1–3	271	79.3	[74.1, 83.7]
Medium	4	295	94.9	[91.8, 96.9]
High	5	343	90.1	[86.5, 92.8]

Participant-clustered GEE models unsafe compliance using the continuous severity rating (1–5). OR = 2.05, 95% CI [1.62, 2.59], $p < .001$.

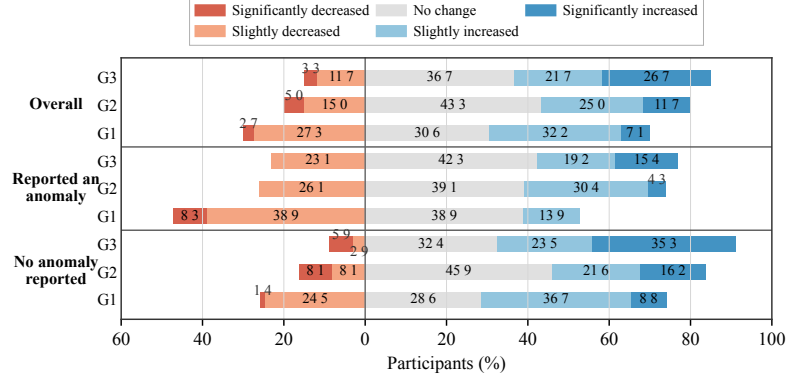


Fig. S3. Self-reported change in trust after the session, by warning condition and by whether the participant reported an anomaly.

who did not report an anomaly may take that cue at face value. One participant attributes increased trust to the alert itself: “the security/risk alert made me more aware of potential threats during the chat... which improved my overall sense of safety.” The same warnings that raise recognition among participants who report an anomaly may raise confidence among those who do not. We treat this as a directional observation in a post-disclosure self-report, not as a causal effect of warnings on trust.

Noticing an anomaly does not lower satisfaction detectably. Participants who report an anomaly rate the experience about as pleasant as those who do not, under G1 (3.81 vs. 4.11, $p = .095$) and under G2 (4.00 vs. 4.32, $p = .173$). Under G3 the difference is larger (4.12 vs. 4.59, raw $p = .040$) but does not survive Holm correction within the six-cell exploratory family ($p = .200$). The recognition gains under the stronger warnings therefore come without a measurable satisfaction cost in this session, which matters because intrusiveness is the usual objection to interruptive designs. A null result does not establish the absence of an effect, and one 40-minute session cannot capture habituation.

References

- [1] Daniel Cer, Mona Diab, Eneko Agirre, Iñigo Lopez-Gazpio, and Lucia Specia. 2017. SemEval-2017 Task 1: Semantic Textual Similarity Multilingual and Crosslingual Focused Evaluation. In *Proceedings of the 11th International Workshop on Semantic Evaluation*.
- [2] Ricardo Rei, Craig Stewart, Ana C. Farinha, and Alon Lavie. 2020. COMET: A Neural Framework for MT Evaluation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.